

Student Result Management System

BY

ABDULLAH NASIR (CT-25076)

M. SAAD AHMED (CT-25089)

FAEEZ UR REHMAN (CT-25090)

INTRODUCTION



This presentation covers a robust system for managing student results. It focuses on how student data is captured, processed, and presented in a clear, structured manner. The goal is to ensure accuracy and transparency in student performance reporting. We will explore key components such as input methods, grade calculation, and final result summarization.

Background



The Student Result Management System is made to make result calculation quick and accurate. In manual result preparation, marks, total, percentage, and grade are calculated by hand which can cause mistakes. This system takes student details and subject marks as input and automatically calculates percentage and grade. It also generates the final result in a clear and neat format.

This saves time and reduces human errors.

The system is easy to use and can handle multiple students.

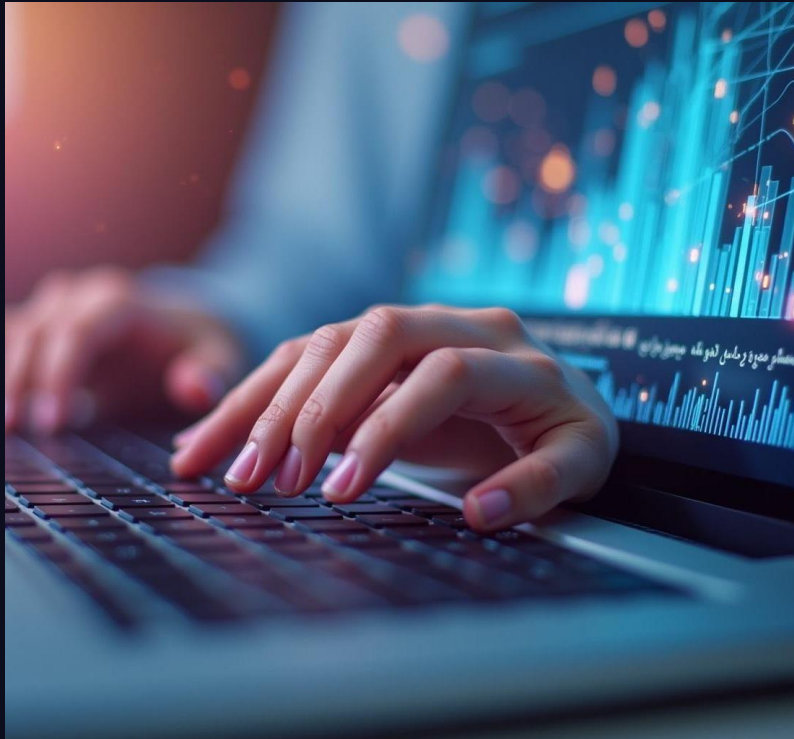
It improves efficiency and makes result management more organized.

GETGRADE FUNCTION

```
// -----  
// Function: getGrade  
// Purpose : Determine grade based on percentage value  
// Input  : float percentage  
// Output : grade character (A, B, C, D, F)  
// -----  
char getGrade(float percentage) {  
    if (percentage >= 80)  
        return 'A';  
    else if (percentage >= 70)  
        return 'B';  
    else if (percentage >= 60)  
        return 'C';  
    else if (percentage >= 50)  
        return 'D';  
    else  
        return 'F';  
}
```

The function `getGrade(float percentage)` determines a student's grade based on their percentage. It uses conditional checks to compare the percentage against defined ranges. If the percentage is 80 or above, it returns 'A'; 70–79 returns 'B'; 60–69 returns 'C'; 50–59 returns 'D'; and anything below 50 returns 'F'. This function helps classify performance into standard grade levels,

INPUT METHODS FOR STUDENT NAME



Accurate entry of the *student's full name* is essential for personalized result management. This system uses an input method that captures names containing spaces without truncation, enhancing data integrity and identification accuracy.

The approach ensures that all student names are recorded correctly for official documentation.

HANDLING MULTIPLE SUBJECTS

```
// ***** Loop to Input Subject Names & Marks *****  
for (i = 0; i < subjects; i++) {  
  
    printf("\nEnter name of subject %d: ", i + 1);  
    scanf("%s", subName[i]); // Only single word names allowed  
  
    printf("Enter marks in %s (out of 100): ", subName[i]);  
    scanf("%f", &marks[i]);  
  
    // Validating Marks (no negative or >100 allowed)  
    if (marks[i] < 0 || marks[i] > 100) {  
        printf("Invalid Marks! Please enter again.\n");  
        i--; // Repeat this iteration  
        continue;  
    }  
  
    obtainedTotal += marks[i]; // Add to total obtained marks  
}
```

The system efficiently handles multiple subjects by using arrays to store subject names and marks, and iterating through them using a for loop. This flexibility supports different academic programs with varying subject counts while maintaining data consistency. It collects each subject name and marks individually and includes validation to ensure marks are not less than 0 or greater than 100, preventing incorrect data entry. This careful input handling sets the stage for accurate percentage calculation and reliable result reporting.

Calculating percentages and grades



Percentages are calculated by dividing obtained marks by the maximum possible marks, then multiplying by 100. Grades are assigned based on predefined percentage ranges to standardize evaluation criteria. This calculation method supports consistent and fair assessment across all subjects. The **getGrade ()** function is used to automatically select the appropriate grade based on the percentage, ensuring accuracy and uniform grading. It enables clear differentiation of student performance levels through standardized grading.

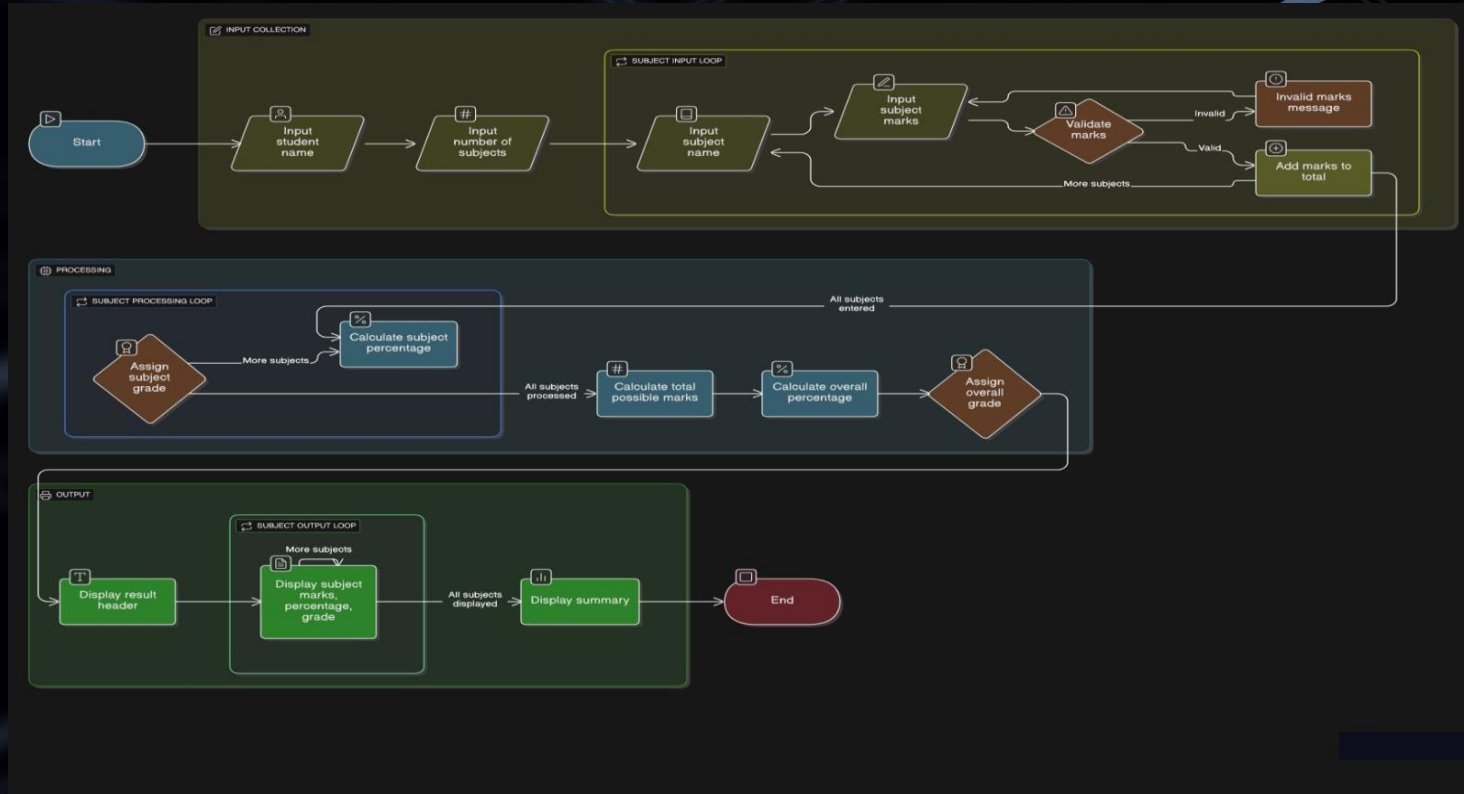
SUMMARY OF TOTAL AND OVERALL PERFORMANCE

The system aggregates marks across all subjects to compute total obtained marks, maximum possible marks, and overall percentage.

An overall grade is assigned to represent the student's comprehensive academic performance. This summary offers a holistic view of the student's achievement in a concise format.

It supports strategic academic decisions and reporting with reliable metrics.

FLOWCHART



Output Of Code

```
Enter Student Name:Hassan Khan
Enter the number of Subjects:4
Enter Name of Subject 1:Maths
Enter Marks in Maths(out of 100):85
Enter Name of Subject 2:English
Enter Marks in English(out of 100):72
Enter Name of Subject 3:Computer
Enter Marks in Computer(out of 100):91
Enter Name of Subject 4:Physics
Enter Marks in Physics(out of 100):86

=====STUDENT RESULT=====
Student Name:'Hassan Khan'
-----
Subject           Marks      Percentage      Grade
Maths             85.00      85.00%          A
English           72.00      72.00%          B
Computer          91.00      91.00%          A
Physics           86.00      86.00%          A
-----
Total obtained marks:334.00
Total subject marks:400.00
overall percentage:83.50%
overall grade:'A'
-----
```



CONCLUSION

This student result management system integrates reliable data capture, validation, and presentation to deliver accurate academic assessments. The structured approach to calculating percentages and grades ensures fairness and transparency. Clear formatting and summarization enhance user understanding and facilitate informed decision-making by stakeholders.

